Samira Pintado

Unit#4 1) rpm = rotations per second × 60

$$rpm = \frac{1}{1.33} \times 60 = \frac{60}{1.33} = 45.11\ rpm$$

$$w = \frac{\text{radians per rotation}}{\text{time per rotation}} = \frac{2\pi}{1.33} = 4.72\ \frac{radians}{sec}$$

answer: B

2) $a_c = w^2 r$

answer: B

3) $a_c = w^2 r$

$100 a_c = (w)^2 r$

$100 = \left(\frac{w^2}{w}\right) = \sqrt{100} = 10$ answer -> D

4) $F_c = m a_c = m w^2 r$

$m = 10g\ (0.01\ kg)$ $w = rpm \times \frac{2\pi}{60}$

$r = 12\ cm\ (0.12 m)$

$w = 200\left(\frac{2\pi}{60}\right) = 20.94\ rad/s$

$a_c = w^2 r = (20.94)^2 \times 0.12$

$a_c = 438.5 \times 0.12 = 52.62\ m/s^2$ answer: $\boxed{B}$

$F_c = m a_c = 0.01 \times 52.62 = 0.53 N$

5. $v = \omega r$ $\qquad$ $v = 30 \cdot 1.4$

$\omega = 30$ rad/s $\qquad$ $v = 42$ m/s

$r = 1.40$ m

answer : C

Unit #4 $\quad$ 1. $W = \vec{F} \cdot \Delta\vec{x} = |\vec{F}||\Delta\vec{x}| \cos\theta$

$W = F_x \cdot \Delta x + F_y \cdot \Delta y$

$W = F_x \cdot \Delta x = 40 \cdot 5 = 200 N \cdot m$

$$\cos\theta = \frac{\vec{F} \cdot \Delta\vec{x}}{|\vec{F}||\Delta\vec{x}|}$$

$|\vec{F}| = \sqrt{(40)^2 + (-30)^2} = \sqrt{1600 + 900} = \sqrt{2500} = 50N$

$|\Delta\vec{x}| = 5m$

$$\cos\theta = \frac{200}{5 \cdot 50} = 0.8$$

$\theta = \cos^{-1}(0.8) = 37°$

Samira Pintado
Midterm 2

Unit #4

1. B
2. B
3. D
4. B
5. C

Unit #5

1. D

2. a. $W = F \cdot d$

$F = 1000N$
$d = 10cm (0.1m)$

$W = 1000 \cdot 0.1 = 100 J$

b. $\Delta KE = W = 100J$

c. $KE = \frac{1}{2}mv^2 = v^2 = 2KE/m$

$m = 0.15kg$
$KE = 100J$

$v^2 = \frac{2 \cdot 100}{0.15} = \frac{200}{0.15}$

$V = \sqrt{1333.33}$

$V = 36.5 m/s$

d. $R = \frac{v^2 \sin(2\theta)}{g}$

$v = 36.5 m/s$
$\sin(2\theta) = 1$
$g = 9.8 m/s^2$

$R = \frac{(36.5)^2 \cdot 1}{9.8} = \frac{1332.25}{9.8} = 136 m$

3. $W = F_{Friction} \cdot d$

a) $F_F = \mu K \cdot F_n$
$\mu K = 0.05$
$F_n = m \cdot g = 40(9.8) = 392N$
$d = 40m$

$F_F = 0.05 \cdot 392 = 19.6N$

$W = 19.6 \times 40 = 784 J$

b) $P = W/t$
$W = 784 J$
$t = 30s$

$P = \frac{784}{30} = 26.13 W$

4. $\text{Power in kW} = \frac{\text{Power in kw}}{1000} = \frac{3.00}{1000} = 0.003 kW$

Energy = Power × Timer

Time = 365(days) × 24(hrs) = 8760 hrs

Enery = 0.003 kW × 8760 hrs = 26.28 kWh

Cost = 26.28 kWh × 0.09 \$/kWh = \$2.36

answer = $\boxed{D}$

$p = mw$

Unit #6

① $m = 20 \times 10^{-25}$ kg

$V1 = 350$ m/s

$V2 = -350$ m/s

$p_{total} = mv1 + mv2 = (20 \times 10^{-25} \cdot 350) + (20 \times 10^{-25} \cdot -350)$

$p_{total} = 20 \times 10^{-25} (350 - 350) = 0$

answer = 0 m/s

② P1:

$m = 1$ kg

$v = 4$ m/s

$px = 4 \times \cos 45 = 2.83$ kg.m/s

$py = 4 \sin 45 = 2.83$ kg.m/s

P2:

$m = 2$ kg

$v = 3$ m/s

$px = 3 \cos 45 = 4.24$ kg m/s

$py = 3 \sin 45 = 4.24$ kg m/s

Px total: $2.83 + 4.24 = 7.07$ kg m/s

Py total $= 2.83 + 4.24 = 7.07$ kg m/s

P total $= \sqrt{(7.07^2 + 7.07^2)} = 10$ kg m/s

Total m $= 1$ kg $+ 2$ kg $= 3$ kg

$V_F =$ Ptotal / m total

$V_F = 10 \div 3 = 3.33$ m/s

3. a) $J = F\Delta t$ $J = 4000N \times 0.200s$

$F = 4000N$ $J = 800 N.s$

$\Delta t = 0.200$

b) $J = \Delta P = m(v_F - v_i)$

$m = 200kg$

$v_i = 2.80 m/s$

$v_F = ?$

$800 N.s = 200 kg (v_F - 2.8 m/s)$

$v_F - 2.8 m/s = \frac{800 N.s}{200 kg} = 4 m/s$

$v_i = 4 m/s + 2.8 m/s = \underline{6.8 m/s}$

4. The example is an <u>elastic collision</u> because there is conservation of both momentum and kinetic energy.

5. $m_{car} = 30{,}000 kg$ $m_{metal} = 110{,}000 kg$

$V_{i\,car} = 0.85 m/s$ $V_{metal} = 0 m/s$

$m_{car} \times V_{car} + m_{metal} \times v_{metal} = (m_{car} + m_{metal}) V_F$

$30{,}000 \times 0.85 + 110{,}000 \times 0 = (130{,}000 + 110{,}000) V_F$

$25{,}500 = 140{,}000\, V_F$

$V_F = \frac{25{,}500}{140{,}000}$

$V_F = 0.182 m/s$

Un

b. $KE_i = 1/2\ m_{car}\ v^2_{car}$

$KE_i = \frac{1}{2} \times 30{,}000\ (0.85)^2 = \frac{1}{2} \times 30{,}000 \times 0.7225$

$= 10{,}837.5\,J$

$KE_F = \frac{1}{2}\ (m_{car} + m_{metal})\ v^2_F$

$KE_F = \frac{1}{2} \times 140{,}000\ (0.182)^2$

$KE_F = \frac{1}{2} \times 140{,}000 \times 0.0331 = 2{,}317\,J$

$KE_{lost} = KE_i - KE_F$

$KE_{lost} = 10{,}837.5\,J - 2{,}317\,J$

$KE_{lost} = 8{,}520.5\,J$

Unit #7

$T = F \times d$

$m_{bricks} = 40kg$ $m_{concrete} = 60kg$

$d_{bricks} = 3\,m$ $d_{concrete} = 2m$

$T_{bricks} = m_m \cdot g \cdot d_b = 40kg \times 9.8 m/s^2 \times 3m = 1176Nm$

$T_{concrete} = m_c \cdot g \cdot d_c = 60kg \times 9.8m/s^2 \times 2\,m = 1176Nm$

A: remain motionless

$m = 20kg$ $W = mg = 20 \times 9.8 = 196N$

$F1 = x = 2m$

$F2 = x = 0.5m$

$F2 = F1 + 196N$

torque equation: $F1(2m - 0.5m) = W(1m - 0.5m)$

$F1 \times 1.5m = 196N \times 0.5m$

$F1 \times 1.5 = 98$

$F1 = 98 / 1.5$

$F1 = 65.33N$

$F2 = F1 + 196N$

$F2 = 65.33N + 196N = 261.33N$

Unit#8 a) $I = 21MR^2$

$m = 1kg$

$R = 8cm = 0.08m$

$\alpha = 2rad/s^2$

$I = 21MR^2$

$I = 21 \times 1kg \times (0.08m)^2$

$I = 21 \times 1 \times 0.00064m^2$

$I = 0.1344 kg.m^2$

b) $T = I.\alpha$

$T = 0.1344 Kg m^2 \times 2rad/s^2$

$T = 0.2688 Nm$

2) a. $\tau = r \times F$

$r = -0.20\hat{i} + 0.20\hat{j}$ m
$F = 12\hat{i}$ N

$\tau = (-0.20\hat{i} + 0.20\hat{j}$ m$) \times 12\hat{i}$ N

$\tau = -2.4\hat{k}$ Nm

b. $\tau = r \times F$

$\tau = -0.20 \times F_y = 2.4$ Nm

$-0.20_m \times F_y = 2.4$ Nm

$F_y = \dfrac{2.4\ \text{N}\cancel{\text{m}}}{0.20\ \cancel{\text{m}}}$

$F_y = -12$ N or 12 N $\downarrow$